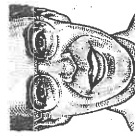


STREETWISE | By James Mackintosh

# Kicking the Tires Of Market Anomalies



Tie together an algorithm, an exchange-traded fund and an academic study finding an anomaly in the markets, and voila! You have a formula for making money. The trouble is, it turns out that most of the supposed anomalies academics have identified don't exist, or are too small to matter.

A new study making waves in quantitative finance tested 447 anomalies identified by academics and found more than eight out of 10 vanish when rigorous tests are applied. Among those failing to reach statistical significance: one anomaly recently set out by the godfathers of quantitative finance, Nobel-winning economist Eugene Fama and his colleague Kenneth French.

The study, "Replicating Anomalies," published this week by Kewei Hou and Lu Zhang at Ohio State University and Chen Xue at the University of Cincinnati, is the biggest test of examples of inefficient markets carried out so far. The trio applied consistent analysis to the supposed anomalies, used the same database of stocks and set higher standards for

Market anomalies that passed the new study's tests included several of the biggest. Cheap stocks indeed beat expensive ones; share prices have momentum; companies that invest a lot underperform, and quality of

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earnings matters. Known as value, momentum, investment and quality, these have become the biggest of the so-called "smart beta" ETFs sucking in tens of billions of dollars.

A lot depends on exactly how the factors are implemented, though, and the researchers dismissed one of the industry-standard Fama-French factors as statistically insignificant: Companies with high operating return on equity don't outperform meaningfully on their tests. Other measures of return on equity did outperform sufficiently, however, underlining the sensitivity of some factors to exactly how they are defined.

One lesson for investors is to be careful about trying to make money by repeating what seems to have worked in the past. If it was so easy, everyone would do it and it would stop working.

A former student of Mr. Fama, Cliff Asness, founder of quantitative hedge-fund manager AQR Capital Management, said he tries to avoid being caught out by

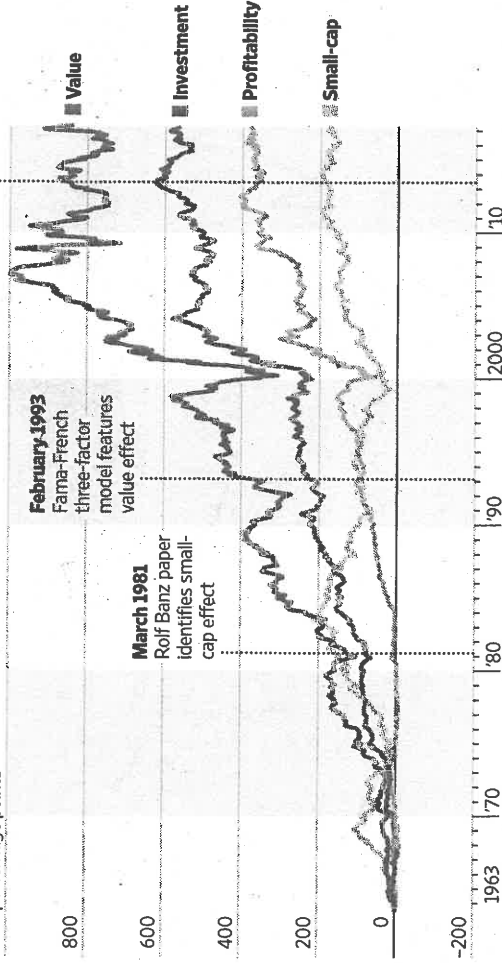
false findings by trading on anomalies he can explain, economically or through investor behavior. To assess whether the market anomalies will continue, he looks for ones which carried on after being identified, can be

## Factors to Beat the U.S. Market

Academics often identify stock anomalies that have beaten the market in the past. Many were simply wrong, according to a new study, but some stopped working after publication and others turned out to be weaker than thought.

### Factor cumulative outperformance\*

1000 percentage points



\*Performance of long-short portfolios on each factor. Investment factor is long companies which invest little, short companies which invest a lot. Note: Through March 31. Source: Kenneth French

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seen in other markets or asset classes, and where minor changes to how they are defined don't much affect the result. These include most famously value, momentum and corporate quality, among others.

Still, he worries that the "awesome effort" in the new paper might lead some to overreact and reject all factors, even those which Messrs. Hou, Xue and Zhang found evidence for.

"Many factors are demonstrably silly, or are highly

correlated versions of the same idea," he said. "Where I get worried is about overreaction [to the paper] and the cynicism it breeds."

Investors are still likely to be confused. There are well over 100 value and high-dividend ETFs in the U.S. alone, tracking large, small or mid-size stocks, based on different definitions and often combined with other factors such as momentum, quality or low volatility. Intelligently choosing between them would mean examining how

indexes are constructed and comparing to the long-term academic studies to see which methodology was best; in practice, for most investors, there is little more to go on than a few years of performance data and fees.

Messrs. Hou, Xue and Zhang provide a handy dismissal of factors that didn't even work that well in the past. But ultimately no one knows whether even previously robust factors like value and momentum will keep working.